A. King Keykhosrow's Mystery

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: brute force, chinese remainder theorem, math, number theory |
https://codeforces.com/problemset/problem/2034/A

There is a tale about the wise King Keykhosrow who owned a grand treasury filled with treasures from across the Persian Empire. However, to prevent theft and ensure the safety of his wealth, King Keykhosrow's vault was sealed with a magical lock that could only be opened by solving a riddle.



The riddle involves two sacred numbers a and b. To unlock the vault, the challenger must determine the smallest key number m that satisfies two conditions:

- m must be greater than or equal to at least one of a and b.
- The remainder when m is divided by a must be equal to the remainder when m is divided by b.

Only by finding the smallest correct value of m can one unlock the vault and access the legendary treasures!

Input

The first line of the input contains an integer t ($1 \le t \le 100$), the number of test cases.

Each test case consists of a single line containing two integers a and b ($1 \le a, b \le 1000$).

Output

For each test case, print the smallest integer m that satisfies the conditions above.

Example

Input
2 4 6 472 896

Output
12 52864

Note

In the first test case, you can see that:

- 4 mod 4 = 0 but 4 mod 6 = 4
- 5 mod 4 = 1 but 5 mod 6 = 5
- 6 mod 4 = 2 but 6 mod 6 = 0
- 7 mod 4 = 3 but 7 mod 6 = 1
- 8 mod 4 = 0 but 8 mod 6 = 2
- 9 mod 4 = 1 but 9 mod 6 = 3
- 10 mod 4 = 2 but 10 mod 6 = 4
- 11 mod 4 = 3 but 11 mod 6 = 5

so no integer less than 12 satisfies the desired properties.